INTRODUCTION AND RATIONALE

Viruses play a multitrophic role in the cycling of marine dissolved organic matter (DOM, Figure 1). Through lysing cells, viruses mediate the global flux of organic matter through the dissolved pool—the **'viral shunt**' (Fuhrman, 1999; Wilhelm and Suttle, 1999)—a flux that has been estimated to be ~150 Gt carbon per year (Lara et al., 2017; Suttle, 2005). Furthermore, viral lysis stimulates phytoplankton and bacterial production of transparent exopolymeric particles (TEP) that aggregate with cell debris, living microbes, and fecal pellets to form 'marine snow' enhancing carbon flow through the biological pump—'**viral shuttle**' (Guidi et al., 2016; Kaneko et al., 2021; Peduzzi and Weinbauer, 1993; Proctor and Fuhrman, 1991).

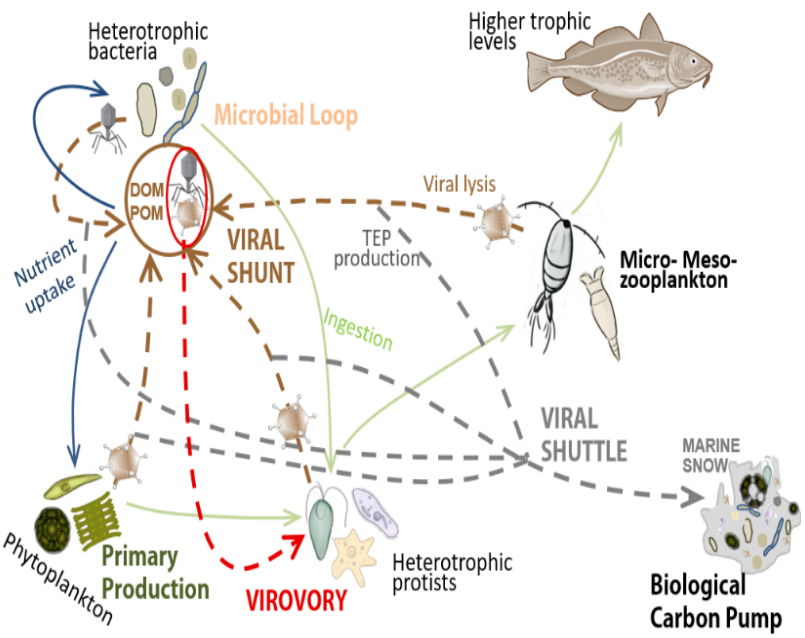

Figure 1. Schematic diagram showing critical roles of virus in nutrients and energy transfer in the ocean

Explicit inclusion of viruses in biogeochemical and ecosystem models is challenging. The vast diversity of hosts and viruses is difficult to capture in models, and we lack appropriate environmental data to test and constrain descriptions of viruses in models. The majority of research has focused on host mortality and virus production via lytic infections, which are estimated to kill 20-30% of the ocean's single-celled organisms daily (Suttle, 1994; Vincent et al., 2021), effectively modulating microbial community dynamics and productivity (Martínez Martínez et al., 2006; Suttle, 1994). Yet, the rate of virion loss, which over sufficiently long timescales must match production, is just as important as a modeling constraint. In marine systems, virus particle loss is generally attributed to UV-degradation (Noble and Fuhrman, 1997; Suttle and Feng, 1992), enzymatic degradation (Corinaldesi et al., 2010; Noble and Fuhrman, 1999, 1997), and aggregation to particles leading to export from the surface ocean (Corinaldesi et al., 2010; Proctor and Fuhrman, 1991). Increasing evidence indicates that direct consumption of virions by protists—**virovory**—is a potentially dominant virus loss process. Additionally, virion ingestion likely supplements grazers' nutrition and is a significant marine microbial food web pathway that redirects nutrients and energy to upper trophic levels. However, grazer nutritional requirements vary in time and space, depending on factors such as particle size and composition of other available food sources (Brown et al., 2020a; DeLong et al., 2023; Deng et al., 2014; Gonzalez and Suttle, 1993; Hennemuth et al., 2008; Olive et al., 2022; preprint-Martínez Martínez et al., 2024).

In this project, we aim to experimentally quantify virovory using two ecologically relevant model systems (*Emiliania huxleyi* virus and *Roseobacter* phage) and environmental microbial communities employing flow cytometry and state-of-the-art isotope tracing approaches. The empirical data will be used to develop a representation of virovory in ecosystem models, contributing to ongoing efforts to quantify virus sinks and assess the roles of viruses in ocean carbon and nutrient cycling.

BACKGROUND

Virions are highly diverse, taxonomically and morphologically, and the most numerous biological entities in the ocean ranging from 10^6 to 10^8 virions mL^{-1} across space and time (Bergh et al., 1989; Wigington et al., 2016) as a result of the balance between virus production and loss rates. Quantitative understanding of virus decay and removal is critical to constrain infection dynamics in models, which is a first step to quantify the 'virus shunt' and 'virus shuttle'.

Ingestion of virions by sponges (Hadas et al., 2006; Welsh et al., 2020), invertebrate larvae (Welsh et al., 2020), the appendicularian *Oikopleura dioca* (Lawrence et al., 2018; Mayers et al., 2021), and protists (Gonzalez and Suttle, 1993; Hennemuth et al., 2008; Olive et al., 2022; preprint-Martínez Martínez et al., 2024) appears to be a significant mechanism for removing viral particles from the water column. Viruses have a nutritional value, being comprised of carbon (0.05– 0.2 fg C per virion (Jover et al., 2014; Kepner Jr. et al., 1998), proteins, nucleic acids, lipids (some taxa) rich in nitrogen and phosphorus (Jover et al., 2014), and micronutrients such as iron (Bonnain et al., 2016) and zinc (Thomassen et al., 2003). However, for some grazers the nutritional gain from ingesting virions appears insignificant compared to the ingestion of phytoplankton (Hadas et al., 2006; Lawrence et al., 2018; Mayers et al., 2021). For phagotrophic nanoflagellates, which are the most important grazers of bacteria across many aquatic ecosystems (Boenigk and Arndt, 2002) and depend on nutrient-rich prey to meet their metabolic needs and support growth, ingesting virions may represent a larger nutritional contribution (DeLong et al., 2023; Gonzalez and Suttle, 1993; preprint-Martínez Martínez et al., 2024). Early experiments with natural coastal communities and enrichment cultures of flagellates (2–20 μm in size) collected at Port Aransas (Texas) demonstrated ingestion and digestion of fluorescently-labeled bacteria and bacteriophages smaller than 100 nm. The measured grazing and digestion rates differed between nanoflagellates and slower ingestion rates were found for smaller virions. At natural abundances of 10^5–10^6 bacteria cells mL^{-1} and 10^7–10^8 virions mL^{-1} (Wigington et al., 2016), virions were estimated to be a significant source of nutrients representing up to 9% of the C, 14% of the N, and 28% of the P that phagotrophs received from ingesting bacteria (Gonzalez and Suttle, 1993). However, the authors pointed out methodological biases that likely led to underestimated values and suggested that under certain environmental conditions, virions might contribute similar or greater nutrition than bacteria to nanoflagellates. Furthermore, the study did not consider the potential full range of virion nutritional content, which depends on factors such as the capsid size, genome size, and presence or absence of lipid membranes (Jover et al., 2014; Mayers et al., 2023).

Eukaryotic viruses in the phylum *Nucleocytoviricota* (NCV) are characterized by large capsids and dsDNA genomes, comparable to those of bacteria. NCVs infect and lyse globally distributed marine plankton, including many seasonal coastal and open-ocean bloom-formers (Brussaard and Martínez Martínez, 2008), and typically reach abundances of 10^4–10^6 virions mL^{-1} (Highfield et al., 2014; Sheyn et al., 2018). NCVs are found across all aquatic systems (Schulz et al., 2020) and may be enriched in oligotrophic marine waters (Edwards et al., 2021; Farzad et al., 2022). Resource transfer from host to viruses seems to be more efficient the larger the virus particles are (Hinson et al., 2023a). Phytoplankton NCVs likely use all of the host's genomic resources before causing lysis, while ssDNA and RNA viruses may maximize their fitness by lysing host cells before exhausting the available resources (Edwards and Steward, 2018), consequently releasing relatively more nutrients than NCVs to the dissolved pool. NCV's capsids have external fibers and bi-layered lipid membranes (Klose and Rossmann, 2014) made of energy-rich C-rich triacylglycerides, P-rich phospholipids, and N-rich betaine lipids (Maat et al., 2014; Schatz et al., 2017), suggesting a higher nutritional value for predators compared to other dsDNA, RNA, and ssDNA viruses. The average elemental (C, N, P) content in NCV particles has been estimated to be one order of magnitude higher than in bacteriophages (Mayers et al., 2023). Furthermore, the large size of NCVs has been hypothesized to be an evolutionary adaptation to be perceived as prey by their phagotrophic host amoebae (Brandes and Linial, 2019; Moreira and Brochier-Armanet, 2008) and, as a consequence, by other nonhost heterotrophic and mixotrophic protists. For example, the freshwater ciliates *Halteria* sp. (15–35 μm) and *Paramecium bursaria* (80–150 μm) ingest algal chloroviruses (100–200 nm, dsDNA, 290 – 370 Kb, lipid bi-layered membrane (Van Etten et al., 2020)) and in the absence of other food sources, the ingestion of 10^4–10^6 chloroviruses per day supported *Halteria*'s growth at a rate similar to that of other heterotrophic protists on different diets (DeLong et al., 2023). However, there are discrepancies between studies. Single-cell genomics analyses of 1–20 μm protists from a eutrophic coastal site in the Gulf of Maine and an oligotrophic coastal site in the Mediterranean Sea suggested that protists (including Choanozoa, Picozoa, Chlorophyta, Cercozoa, Stramenopiles, Alveolata, and Haptophyta) ingest primarily ssDNA viruses and bacteriophages, and to a lower extent, bacteria and NCV virions. This conclusion was reached from the presence of viral reads in single amplified genomes of nonhost protists. Interestingly, the occurrence of viral sequences was higher in protists from the Gulf of Maine, where every single Choanozoa and Picozoa contained viral sequences (Brown et al., 2020a), revealing potentially significant differences across grazer taxa and ecosystems. A limitation of that study is that single-cell genomics cannot discern ingestion from digestion. Fast digestion of bacteria or NCVs would hinder their detection by this approach. An older study suggested that viruses (<0.2 μm) and viral lysate products, though labile, were not efficiently incorporated into bacterial biomass (Noble and Fuhrman, 1999). However, that study did not investigate ingestion and incorporation by protists. An open question that we will address in this project is whether protists may be more efficient than bacteria at breaking down the structural biopolymers of viruses, particularly NCVs, which may be more palatable due to the reduced, energy-rich carbon within their lipid membranes (Mayers et al., 2023).

To advance our understanding of the ecological and biogeochemical relevance of virovory in marine ecosystems, it is critical that we investigate the bioavailability of different types of viruses and where and when in the ocean virovory is quantitatively important by empirically quantifying rates of ingestion, digestion, and assimilation, and the impact on grazer physiology, e.g., growth rates. While it is not realistic to investigate every virus type, expanding culture-based experiments to representative, ecologically relevant model systems for NCVs and bacteriophages and natural community assemblies over spatiotemporal scales will yield the necessary information to parameterize trophic and biogeochemical models.

PRELIMINARY RESULTS: virion loss and trophic transfer of viral C and N through virovory within a coastal microbial community.

Studies partitioning viral losses between virovory, light damage, and enzymatic breakdown have indicated that grazing by protists may be a variable but relatively minor component of viral loss. For example, filtration of coastal seawater to remove organisms reduced viral removal by 20%, pointing to factors other than virovory as dominant loss processes (Noble and Fuhrman, 1997) while another study predicted that consumption by flagellates may be as low as 1% of total viral losses (Suttle and Feng, 1992). Critically, the majority of prior experiments were conducted with bacteriophage. Given the greater nutritional value of NCVs, their rate of consumption by bacteria and protists may be a larger source of mortality than for smaller bacteriophages. Furthermore, it has been hypothesized that larger viruses may be more stable and therefore resistant to damage by heat shock or osmotic pressure (De Paepe and Taddei, 2006; Edwards et al., 2021). Therefore, we hypothesized that virovory may be a dominant loss process for relatively large viruses.

We performed three independent experiments where we incubated axenic ^{13}C and ^{15}N isotope-labeled *Emiliania huxleyi* viruses (EhVs, NCVs with 160-180 nm diameter (Wilson et al., 2002)) with either microbe-free seawater or seawater including bacteria and protists sampled in different years and seasons at a coastal location in the Gulf of Maine, Atlantic Ocean (preprint-Martínez Martínez et al., 2024). In addition to its ecological relevance, the *Emiliania*-EhV system is at the forefront of marine virus modeling efforts (Hinson et al., 2023a; Nissimov et al., 2019; Thamatrakoln et al., 2019). The incubations were maintained in the dark to test for heterotrophy. Each experimental design was improved using information obtained from the previous experiment. For example, in the first experiment, we examined the virus removal in bottles on long timescales (up to 69 days) to get an initial idea about the turnover of viral particles with and without the microbial loop. In this experiment, the EhV-lysates contained dissolved organic matter with molecular weight over 50 KDa. The subsequent experiments ran for 8 days and EhV-lysates purification was enhanced by removing algal-derived DOM less than 300 KDa. In the second experiment, we investigated changes in microbial taxa composition to identify potential virus degrading/ingesting microbes. In the third experiment, we separated the influence of most protists from that of bacteria by adding another size fraction to the set of treatments. Using flow cytometry enumeration and nanoSIMs isotope measurements we quantified the loss of viruses due to abiotic vs biotic factors and the flux of viral C and N into protists and bacteria.

The flow cytometry data showed that the presence of bacteria and protists, combined (light grey line, experiments 1 & 2) and individually (light and dark grey lines, experiment 3), distinctly influenced the removal of viruses (Figure 2 top). We found that by eliminating all, or at least most, of the cellular organisms (<0.2 µm or <0.1 µm, black lines), roughly 50% of the viruses were removed through mostly abiotic processes by the end of the experiments. Adding biotic processes (inclusion of bacteria and protists <100 µm, grey lines) led to a total removal of 83% to 96% of the viruses. Additionally, in experiment 3, we found that bacteria and protists were responsible for increasing the daily removal rate of viruses by 33% (<1.2 µm, bacteria-enriched filtrate) to 85% (<100 µm, bacteria and protists fraction) compared to abiotic processes alone.

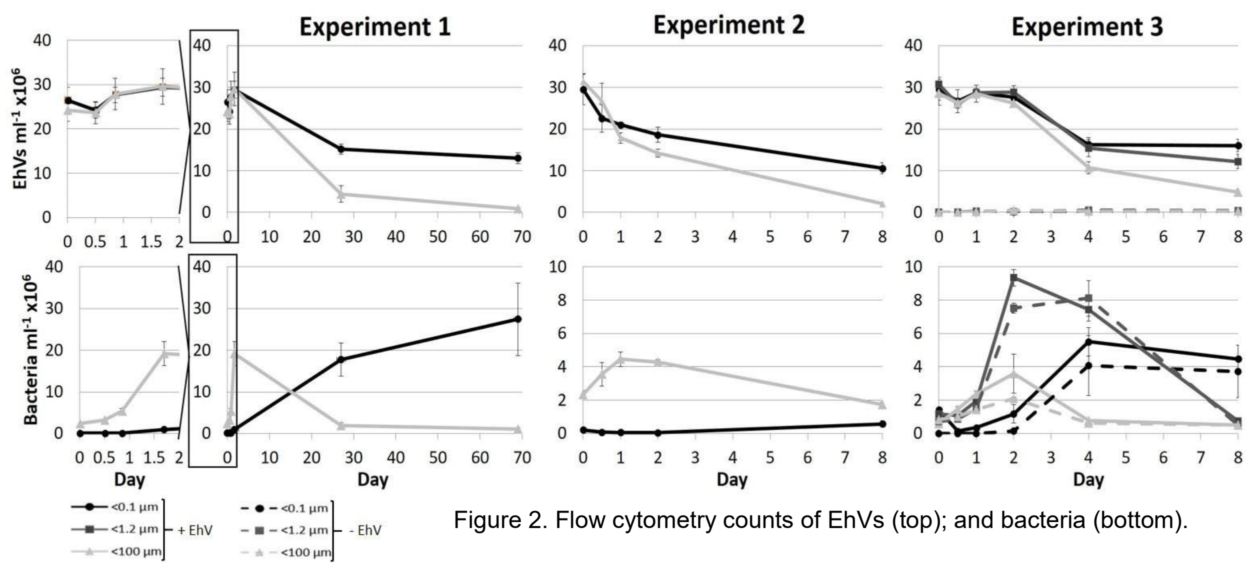

Figure 2. Flow cytometry counts of EhVs (top); and bacteria (bottom).

NanoSIMS isotope tracing showed that the disappearance of EhVs in the incubation flasks was correlated with increasing isotope enrichment in bacteria-size and protist-size cells (Figure 3). We analyzed 27 protist cells of variable sizes and morphologies and found positively correlated C_{net} and N_{net} values (representing the fraction of a cells's biomass originating from the labeled virions), indicating direct incorporation of viral C and N rather than cross-feeding from remineralized N. The high levels of C and N in some protists (Figure 3 D) suggest intense grazing over multiple cell divisions using viral C and N as the sole source of nutrition.

The microbial community in experiment 2 was taxonomically identified and their relative abundances determined by 16S and 18S rRNA sequencing. Bacteria in the families

Campylobacteracea, *Colwelliaceae*, *Oceanospirillaceae*, and *Porticoccaceae*, and heterotrophic eukaryotes within the classes Oligotrichea (ciliates), Picomonadidea (picobiliphyte), and Imbricatea (cercozoa) were the most abundant taxa and showed the largest increases by the end of the experiment (preprint-Martínez Martínez et al., 2024). Consequently, these taxa are the most likely predators of EhVs during that experiment.

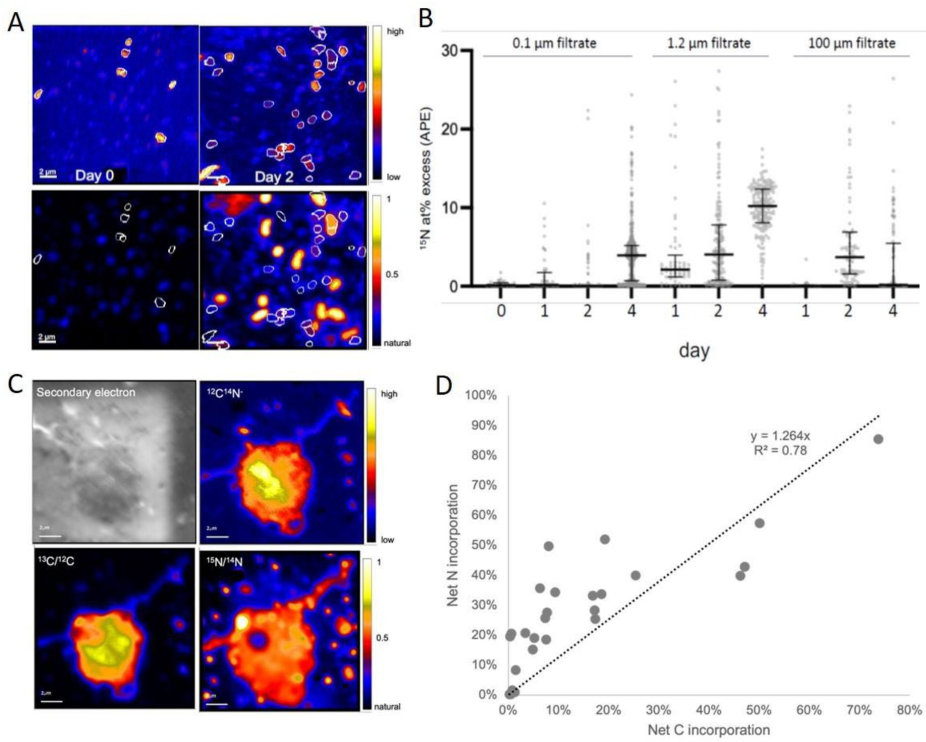

Figure 3. Bacteria-size organic particles from a representative incubation flask (biomass $^{12}C^{14}N$, top panels; isotope enrichment $^{15}N/^{14}N$, lower panels) (**A**) increased in isotope enrichment over time (**A**, **B**). Representative protist cell detected by SEM (black and white) revealed by nanoSIMS to be isotopically enriched (**C**). The 27 analyzed protist cells exhibited variable but positively correlated ^{13}C and ^{15}N enrichment (**D**).

Despite slight alterations to the experimental designs and the sampling time of the year, the data were relatively consistent and provided direct evidence for high EhV loss and the transfer of viral C and N into the microbial loop through protist grazing and bacterial breakdown. Our data highlight the need to investigate virovory further to better understand and predictably model aquatic C and N cycling.

PRELIMINARY RESULTS: food web modeling.

To develop initial quantitative intuition, and initiate a framework for virovory modeling more broadly, we developed a simple food web model to interpret the results in Figure 2. The ordinary differential equation model in Equation 1 allows a population of viruses (V, ml^{-1}) to be preyed upon by heterotrophic bacteria (B, ml^{-1}) and a protist grazer (G, ml^{-1}). In Equation 1, viruses are fed upon by bacteria and grazers following a Holling I functional response with coefficients $\varphi_{g,v}$ and $\varphi_{b,v}$ (both ml day^{-1}), and viruses decay at rate δ_v (day^{-1}). Bacteria grow at rate μ_b (day^{-1}) and consume viruses with efficiency $\varepsilon_{b,v}$ (non-dimensional, n.d.). Bacteria are in turn preyed upon by protists, also following Holling I, at rate $\varphi_{g,b}$ (ml day^{-1}), and we assume bacterial mortality unrelated to protist grazing is captured with a quadratic mortality term, with coefficient δ_b (day^{-1}). Grazers consume viruses and bacteria with efficiency $\varepsilon_{g,b}$ and

$$
\begin{aligned}
\frac{dV}{dt} &= -\underbrace{\varphi_{g,v}GV}_{\substack{virovory\\ by\ grazers}} - \underbrace{\varphi_{b,v}BV}_{\substack{virovory\\ by\ bacteria}} - \underbrace{\delta_v V}_{\substack{viral\\ decay}} \\
\frac{dB}{dt} &= \underbrace{\mu_b B}_{\substack{bacteria\\ growt\ h}} + \underbrace{\varepsilon_{b,v}\varphi_{b,v}BV}_{\substack{virovory\\ by\ bacteria}} - \underbrace{\varphi_{g,b}GB}_{\substack{protist\\ grazing}} - \underbrace{\delta_b B^2}_{\substack{bacterial\\ mortality}} \qquad (1) \\
\frac{dG}{dt} &= \underbrace{\varepsilon_{g,b}\varphi_{g,b}GB}_{\substack{protist\\ grazing}} + \underbrace{\varepsilon_{g,v}\varphi_{g,v}GV}_{\substack{virovory\\ by\ grazers}} - \underbrace{\delta_g G^2}_{\substack{grazer\\ mortality}}
\end{aligned}
$$

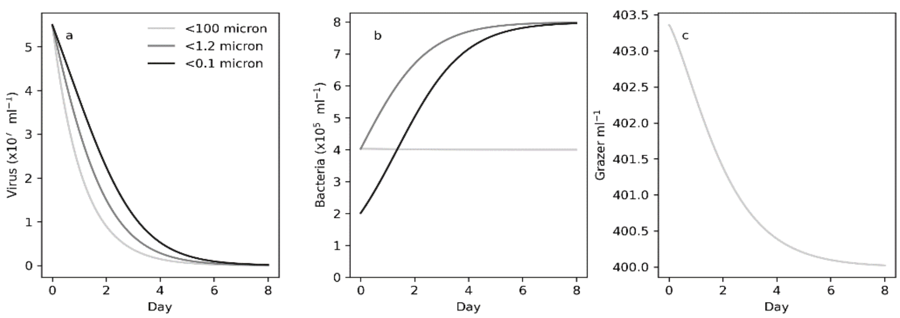

Figure 4. Modeled dynamics with Eq. 1 a) viruses b) bacteria and c) protists.

$\varepsilon_{g,v}$ respectively (both n.d.), and grazer mortality is also quadratic with coefficient δ_g (day^{-1}). In Figure 4, we assess the ability of the model in Equation 1 to capture the dynamics, measured by flow cytometry, for viruses and bacteria in Figure 2. To reproduce filtration by 100, 1.2, and 0.1 µm, we assumed the removal of all microzooplankton larger than 100 µm (light grey lines, Figure 2), protists (dark grey lines, Figure 2), and bacteria (black lines, Figure 2), respectively. The model captures diminishing virions loss with increasingly fine levels of filtration (Figure 4a). It also capture the rise in bacteria with removal of protists in the 1.2, and 0.1 µm treatments (Figure 4b). The model predicts a modest decline in Grazer abundance in the 100 µm due to the nutritional shortfall associated with loss of viruses (Figure 4c).

The model in Equation 1 and Figure 4 has limitations preventing its application to natural ecosystems. For example, there is only one group of viruses, when in practice there are diverse assemblages of virus that will respond differently to manipulation by filtration. Furthermore, we did not account for changes in protist losses associated with the removal of microzooplankton, and the potential this could have to modify bacteria dynamics. Nevertheless, the gradient in mortality associated with filtration in Figure 4a provides a basis on which to develop more realistic models of virovory within a food web context. Our experimental plan is designed to address shortcomings in our preliminary work for inclusion of virovory in ecosystem models.

HYPOTHESES

The premise of this proposal is that viruses play multiple and dynamic roles in the fate of organic matter and the flow of energy in the ocean; yet, the specific virus-driven pathways have not yet been rigorously quantified. Arguably, the least studied of those is virovory. In this project we will test the following specific hypotheses to first establish the significance of virovory as a virion loss process, then address whether different virus types are grazed at different rates and contribute differentially to grazers' nutritional requirements. Our experimental design will inform development of a representation of virovory in a virus-explicit ecosystem model.

Hypothesis 1 (H1): *Virovory is a natural sink of marine viruses that rivals viral loss rates due to UV-degradation, enzymatic breakdown, aggregation, and sinking combined.*

Rationale 1: In our preliminary experiments (below) we measured significantly larger disappearance of virions in incubations with protists (~83-96% removal) than in flasks where removal could only be attributed to bacterial activity or abiotic factors, such as decay or attachment to the experimental flasks' walls (~45-63% removal).

Hypothesis 2 (H2): *Nutrient transfer (normalized to virion abundance and volume) from large enveloped dsDNA eukaryotic viruses (NCVs) is higher than from smaller virus particles.*

Rationale 2: The average elemental (C, N, P) content in NCV particles has been estimated to be one order of magnitude higher than in bacteriophages (Mayers et al., 2023). Additionally, since lipids are energy-rich biomolecules (Amend et al., 2013), the presence of bi-layered lipid membranes potentially increases the energy content of NCVs.

Hypothesis 3 (H3): *Virovory contributes more nutrition for heterotrophic protists in permanently or seasonally oligotrophic waters than in eutrophic marine ecosystems.*

Rationale 3: Phagotrophic nanoflagellates, the most important grazers of picoplankton across many aquatic ecosystems (Boenigk and Arndt, 2002), depend on nutrient-rich prey to meet their metabolic needs and support growth. Variation in N:P regimes reflects selective pressures favoring small plankton with lower P storage capacity in oligotrophic environments, and larger plankton with a higher P storage capacity in nutrient-rich environments (Inomura et al., 2022). In addition to carbon, virions comprise proteins, nucleic acids, and in taxa like NCVs which appear to be enriched in oligotrophic waters (Edwards et al., 2021; Farzad et al., 2022), lipids rich in nitrogen and phosphorus (Jover et al., 2014). Virions may also be a source of important micronutrients such as iron (Bonnain et al., 2016) and zinc (Thomassen et al., 2003).

RESEARCH APPROACH

Mathematical modeling of trophic transfer in ecologically complex aquatic systems is a formidable task. Global food web and biogeochemical models still do not typically resolve viral infection (Dutkiewicz et al., 2009; Stock et al., 2014), although a handful of studies have begun resolving viruses on regional scales (Nissimov et al., 2019; Weitz et al., 2015; Xie et al., 2022). One issue is the lack of appropriate laboratory data to inform model parameterization. Virovory is one example of a process for which, to our knowledge, there are no existing model-data comparisons. Our experimental design and approach integrates mathematical modeling techniques throughout. A sensible starting point to address the above hypotheses is to deconstruct food web complexity. To that end, we are planning experiments with size-fractionated natural microbial communities amended with isotope-labeled clonal viral isolates. We have identified the well-characterized phages of *Sulfitobacter* sp. (*Roseobacter* clade, *Proteobacteria*) (Ankrah et al., 2014; Jover et al., 2014) and *Emiliania huxleyi* (*Prymnesiophyceae*) viruses (Mackinder et al., 2009; Wilson et al., 2005, 2002) as ecologically representative models for our comparative study. To expand the relevance of our work, we plan to collect natural microbial communities at different times of the year from a coastal and an open ocean site representing eutrophic and oligotrophic regimes. The selected virus model systems are cosmopolitan, ecologically relevant, and share morphological characteristics with a broad range of marine bacteriophages and lipid-enveloped Nucleocytoviricota (NCVs) (see Methods, including contingency plans).

Experimental manipulations of marine microbial communities – grazing experiments. We will test ***H1–3*** in parallel through a series of manipulative experiments with environmental microbial communities (eukaryotes and prokaryotes) collected at the Bigelow Laboratory's dock (fall/winter 2025, spring/summer 2026; Maine, shallow coastal, mixed layer, nutrient-rich site) and from the Sargasso Sea at the Bermuda Atlantic Time Series (BATS) site (fall/winter 2026, spring/summer 2027; open ocean, seasonal stratification, oligotrophic) (see letter from Drs. Johnson and Bates). So far, studies of virovory have focused on a single model grazer-virus system or study site, without temporal resolution. By performing experiments with phages and NCVs during seasons of high and low production, we will gain an understanding of differential removal rates (***H1***) and nutrient transfer efficiency (***H2***) for viruses representative of broad, ecologically representative types, in addition to how nutrient availability in the water column influences the dependence on virions as a nutrition source (***H3***). Grazing rates may also be influenced by water temperature (through effects on metabolism) and light conditions (Moeller et al., 2019). Including geographical and temporal sampling representing contrasting environmental conditions will allow extrapolation to other systems and ecosystems.

Prior studies suggest that heterotrophic protists smaller than 40 µm in diameter are likely the primary consumers of virions (Brown et al., 2020b; DeLong et al., 2023; Gonzalez and Suttle, 1993; preprint-Martínez Martínez et al., 2024). Our experimental set up (Figure 5) consists of incubations (in triplicate) of environmental communities size-fractionated between (1) <40 µm, containing protists, prokaryotes, and viruses; (2) <0.8 µm, containing prokaryotes and

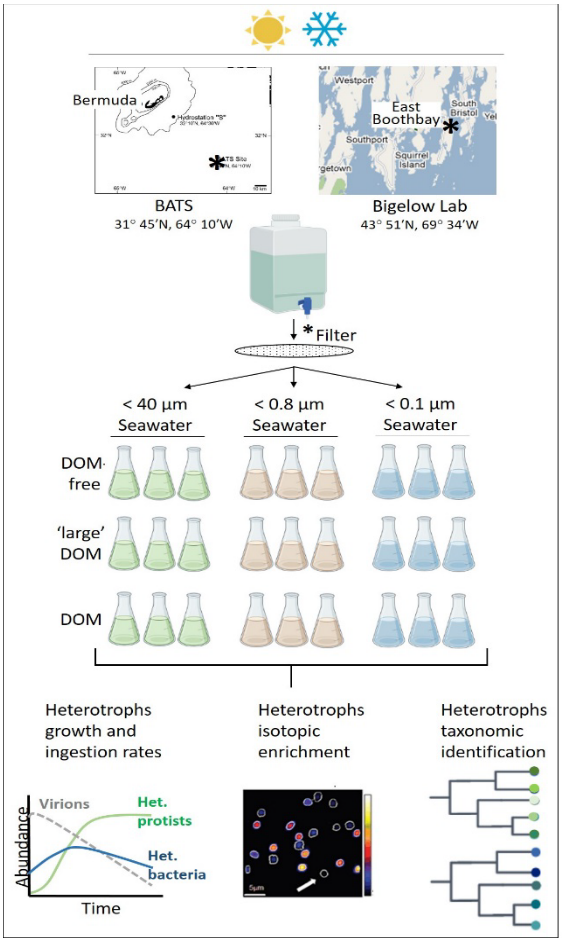

Figure 5. Experimental design to test H1-3

viruses; (3) <0.1 μm, containing only relatively small viruses (i.e., heterotrophs-free control), each amended with either DOM-free artificial seawater, "large"DOM (including virions), or DOM obtained from either mechanically-lysed healthy or virally-lysed cultures grown in modified media with ^{13}C and ^{15}N isotopes (see Methods). The flasks will be incubated at *in situ* temperature in the dark (to limit phototrophic growth) and monitored daily to measure changes in the abundance of virions, heterotrophic bacteria, and heterotrophic protists. Samples will also be collected to determine isotopic enrichment in heterotrophs and taxonomically identify them.

Our approach will also allow resolving how much of the removal of virus particles is biotic (i.e., ingestion by protists, or bacterial degradation), or abiotic (e.g., attachment to the incubation flasks and/or degradation by ectoenzymes present in the water) by comparing changes in virion abundance in the <40 μm, <0.8 μm, and <0.1 μm, respectively **(*H1*)**. Nutrient transfer **(*H2*)** will be tested by comparing growth and isotope incorporation across flasks amended with similar total viral carbon (estimated based on virus abundance and particle volume) in the form of phage or NCV particles. Similarly, we will also test how much nutrition is comparatively gained by protists and prokaryotes at each geographical location and time of the year by measuring population growth rates and ^{13}C and ^{15}N incorporation within each environmental community fraction **(*H3*)**.

Virion loss and heterotroph growth rates (suggesting ingestion and digestion) will be calculated from flow cytometric enumeration of viruses, bacteria, and protists using well-established methods that we routinely employ at Bigelow Laboratory (Brussaard, 2004; Christaki et al., 2011; Marie et al., 1999; Rose et al., 2004). As reported in our preliminary results, we will use nanoSIMS and SEM to provide further evidence of ingestion of isotopically enriched organic matter and/or viruses and estimate the percentage of protist and bacteria C and N contents derived from the isotopically labeled food sources by calculating the ^{13}C and ^{15}N enrichment levels. Furthermore, we will investigate which protistan and/or bacterial taxa benefit from direct ingestion of virions or virally-modified organic matter. Changes in the taxonomic composition (clade level) of bacteria and protists will be investigated by 16S rDNA and 18S rDNA metabarcoding (Blanco-Bercial et al., 2022; preprint-Martínez Martínez et al., 2024; Zemb et al., 2020) in combination with CHIP-SIP analyses (Mayali et al., 2012)(see Methods), which measures the isotope compositon (also with nanoSIMS) of community RNA that is hybridized to a microarray.

Our preliminary work showed that differences in EhVs decay in flasks containing natural microbial communities smaller than 100 μm compared to flasks where (most) organisms larger than 0.1 – 0.2 μm were removed were observable and significant after 2 days and for up to 8 days (Figure 2). Based on those observations, we propose to run the experiments for up to 8 days, with 24-hour sampling intervals. However, daily flow cytometry counting of virions and cells will inform whether the experiments can be terminated earlier or extended. One argument against extending the experiments for too long is to minimize bottle effects on the microbial community and biases against organisms not amenable to conditions in the laboratory. We expect natural bacterial abundances to range between 10^5-10^6 cells mL^{-1} (Wigington et al., 2016) and heterotrophic protist (<40 μm) abundances to range between 10^2-10^4 cells mL^{-1} (Choi et al., 2012; Gonzalez and Suttle, 1993; Lin et al., 2022; Sanders et al., 1992). We will amend the "large" DOM flasks with a final concentration of approximately 1-3 x 10^7 virions mL^{-1}, which is a typical virus abundance in many marine ecosystems (Wigington et al., 2016). DOM amendments will match the quantity corresponding to a lysate with the same virion abundance. We will run the experiments in 1 L volume incubations to allow for adequate daily subsampling without introducing significant overall volume changes. In addition to including triplicates for each treatment, the reproducibility of the results will be confirmed by performing two independent experiments each season at each location within ~one week apart. Testing every single possible combination of factors is experimentally unrealistic. Our research plan considers the investigation of aspects of virovory and microbial interactions that will significantly advance understanding the impact of virovory in marine ecosystems, while maintaining a realistic workload.

Mathematical modeling of virovory. An ordinary differential equation (ODE) model similar to Equation 1 (see preliminary work) will be developed. We will evaluate and test different model formulations, for example exploring different algebraic forms (e.g. Holling II, III (Holling, 1959)). We will explore different model formulations prior to data generation, using preliminary flow cytometry data for guidance, and then use the experimental data to discriminate between candidate models. Working in a step-wise manner, we will fit different versions of the model to experimental data. Where possible, parameters will be retrieved by directly fitting the model with time-series data. In some instances, parameters will not be quantifiable directly with fitting, due to non-identifiability (Dong et al., 2021). Our fitting technique (see Methods) is designed specifically to identify such instances. When parameters cannot be retrieved through fitting, we will conduct literature searches to identify whether parameters can be constrained or inferred with existing information. In some instances, it may not be possible to uniquely identify parameters. For example, it may not be possible to disentangle efficiency of virovory by bacteria $\varepsilon_{b,v}$ from the cell-specific rate of virovory $\varphi_{b,v}$. In such instances, simplifying model assumptions will be made, in this example by combining these two parameters into a single model parameter.

Models will be formulated in abundance units, to be compatible with our flow-cytometry data. However, with estimates of elemental quotas for phytoplankton and protists (Menden-Deuer and Lessard, 2000), along with estimates of virus elemental composition (Jover et al., 2014; Mayers et al., 2023), we will convert from abundances to elemental concentration. Predictions of elemental concentration and transfer will be directly comparable with our ^{13}C and ^{15}N nanoSIMS incorporation data. Comparison of model predictions of elemental flow with nanoSIMS data in contrasting trophic conditions will provide an additional assessment of ***H3***, and an indication that our model is appropriate to incorporate flow-cytometry and nanoSIMS measurements into food web and biogeochemical models.

We will fit our model of virovory (e.g. Equation 1) to experimental data describing dynamics of small (phage) and large (NCV) viruses introduced to natural communities of bacteria and protists. To test ***H2***, we will ask whether parameters describing rates of consumption are higher for NCVs than for phages. In Equation 1, these parameters are $\varphi_{b,v}$ and $\varphi_{g,v}$ for bacteria and protists, respectively. Our approach to model fitting (see methods) will provide estimates of all parameters as well as probability distributions quantifying uncertainty regarding experimental and biological error. We will test for differences in probability distribution of each parameter (e.g., using Kolmogorov-Smirnov) to evaluate the results' statistical significance.

Model parameters describing bacteria and protist-specific rates of virovory ($\varphi_{b,v}$ and $\varphi_{g,v}$, respectively) combined with abundances of protists and bacteria, will provide estimates of rates of loss due to predation by bacteria and protists. To test ***H1**, w*e will compare these rates with literature data for comparable values of light degradation (e.g., (Wilhelm et al., 1998)).

Projecting the impacts of virovory with a microbial ecosystem model. The Darwin ecosystem model (Dutkiewicz et al., 2020; Follows et al., 2007) has been used to represent the effects of *E. huxleyi* viruses on host populations in the North Atlantic (Nissimov et al., 2019). To overcome the inherent disconnect between our bottle and culture experiments and natural ecosystems, we will embed a description of virovory in the Darwin model and evaluate the impact on *E. huxleyi* populations and their viruses in the North Atlantic. The goal will be to understand if our virovory measurements lead to meaningful impacts on the distribution of *E. huxleyi* cells and their viruses, as a step toward testable predictions of virovory on much larger scales in the ocean.

METHODS DESCRIPTION

Host-virus systems selection. We recognize the importance of microbial diversity, but it is not possible to explore every marine system. Instead, we will select the marine bacterium *Sulfitobacter* sp. (*Roseobacter* lineage) and the phytoplankton *Emiliania huxleyi* (Prymnesiophyceae) and their associated viruses (Ankrah et al., 2014; Jover et al., 2014; Mackinder et al., 2009; Wilson et al., 2005, 2002), which are established in culture, ecologically relevant, and represent diverse systems. For example, We currently maintain and have experience working with *E. huxleyi* and their viruses (EhVs). We will obtain the *Sulfitobacter*-phage system and guidance on maintenance from Dr. Buchan (UTK) (see letter).

Members of the *Roseobacter* clade, including the genus *Sulfitobacter*, typically comprise over 15% of coastal and mixed-layer ocean bacterial communities (Buchan et al., 2005). *Sulfitobacter* spp. are readily cultivated and are, like other *Roseobacter* genera, commonly used as a foundation model in bacterial ecology and physiology studies. As a group, they are responsible of critical biogeochemical transformations (Buchan et al., 2005), which in turn, are impacted by phage infection (Ankrah et al., 2014; Jover et al., 2014).

E. huxleyi is globally distributed and the most numerous coccolithophorid (calcifying phytoplankton) in the ocean. *E. huxleyi* is found offshore in the Gulf of Maine (Balch et al., 1991) and the Sargasso Sea (Haidar and Thierstein, 2001). While an *E. huxleyi* diet can support growth of heterotrophic protists (Anderson and Menden-Deuer, 2017; Goode et al., 2019; Harvey et al., 2015), grazing pressure on *E. huxleyi*-dominated phytoplankton appears to be generally low (Fileman et al., 2002; Olson and Strom, 2002). Instead, *E. huxleyi*-dominated bloom termination appears to be intrinsically linked to lytic viral infection resulting in the accumulation of large quantities of large virus particles in the water column (Bratbak et al., 1993; Brussaard et al., 1996; Martínez Martínez et al., 2012), which we hypothesize represent a significant source of nutrition to microbial communities, mainly heterotrophic protists.

One caveat of using EhVs and *Sulfitobacter*-phage is the possibility that susceptible hosts are present at the time we collect seawater at the selected sites. Infection and virus propagation would complicate interpreting the results. While incubating the experiments with natural communities in the dark would minimize phytoplankton growth and virus propagation, that is not sufficient to prevent potential infection of heterotrophic *Sulfitobacter* spp. Consequently, before starting our experiments, we will evaluate the need for inactivating the virus and phage particles in the lysates following simple techniques that use heat, UV-radiation, and/or mild detergents (Elveborg et al., 2022). Alternatively, if inactivation is unsuccessful in preliminary tests, or if it leads to the loss of the lipid envelope in EhVs or capsid breakage as detected by flow cytometry, we will use surrogate non-marine viruses such as chloroviruses (NCV) (DeLong et al., 2023) (see letter from Dr. Dunigan) and *E. coli* phage T4 (available at the ATCC collection).

Culture maintenance and preparation of isotope-labeled cultures and DOM fractions. Axenic *E. huxleyi* strain CCMP374 will be maintained in artificial seawater-based f/2-Si media (Guillard, 1975) at 16 °C under a 14:10 h light:dark cycle at ~100 μmol photons $m^{-2}s^{-1}$ (Gilg et al., 2016). The two genetically similar host-prophage systems *Sulfitobacter* sp. strains CB-A and CB-D will be maintained in artificial seawater-based Roseobacter Marine Media supplemented with either 4 mM L-glutamic acid [glutamate] or 10 mM sodium acetate [acetate] (Basso et al., 2022) at 25°C in the dark with 200 r.p.m. agitation (Ankrah et al., 2014). Specific media and culture conditions will be used for alternative systems (e.g., *Chlorella* spp. and *E. coli*), if needed. Using artificial seawater-based media allows better control over nutrient concentrations across experiments.

Isotope labeling will be achieved by transferring the host cultures into modified media with ^{13}C and ^{15}N isotopes (Ankrah et al., 2014; preprint-Martínez Martínez et al., 2024). When the *E. huxleyi* cultures reach mid-exponential growth phase (~1 × 10^6 cells ml^{-1}) they will be split into two equal volumes. One of those volumes will be inoculated with the EhV-86 strain at a virus:host cell ratio of approximately 30 (Gilg et al., 2016), using a fresh virus stock. The other aliquot will be kept as virus-free control. All the flasks will be maintained under normal culture conditions until culture clearance in the infected flasks, typically 3-6 days. *Sulfitobacter* spp. cultures will be grown to an OD_{540} ~0.17. Under these conditions, spontaneous phophage induction and cell lysis begins to be detectable in CB-A, but not in CB-D, consequently, CB-D will be used as control. Parallel *Sulfitobacter* sp. strains CB-A and CB-D will be maintained until significant prophage induction and lysis is observed in CB-A. Host and virus particle abundances will be measured at the beginning and the end of the experiments by flow cytometry (Brussaard, 2004; Marie et al., 1999). Alternatively, quantitative PCR can be used for enumerating CB-A phages (Ankrah et al., 2014) and EhVs (Pagarete et al., 2009). Host cells and virus particles enumeration will allow normalizing additions in the grazing experiments.

The control cultures will be lysed by sonication. The lysates from the mechanically and virally lysed cultures will then be size fractionated to obtain the organic matter fractions for the grazing experiments. DOM is generally accepted as the fraction of total organic matter (TOM) passing through filters with pore sizes in the range of 0.22 – 0.7 µm. For the purpose of this study, we operationally define particulate organic matter (POM) as the fraction of TOM retained by GF-75 filters (0.3 µm nominal pore size, Advantec MFS, USA) (Halewood et al., 2022), DOM as the fraction passing through a 300 KDa molecular weight cutoff membrane, and "large" DOM as the fraction in between. Preliminary work (see above) confirmed that EhV particles (160–200 nm diameter) are retained and concentrated by tangential flow filtration (TFF) of lysates using a 300 KDa MWCO cartridge (35 nm nominal pore size). It is realistic to expect this approach is also adequate for phages with capsid diameters of at least 70 nm (e.g., *Sulfitobacter* CB-A)**.**

Sampling of environmental communities for grazing experiments. For each grazing experiment we will collect 30 L of seawater at a depth of ~0.5 m below the surface. For the experiments at Bigelow Laboratory, we will collect water off our dock using a hand-operated Niskin bottle. Ancillary oceanographic data (temperature, salinity, oxygen concentration, and pH) will be obtained using a YSI data logger to provide hydrographic context. The water will be processed (size fractionated through a 40 µm and 0.8 µm and 0.1 µm filters) and the experiments set up within 4 hours from arriving to the laboratory. We will leverage BATS cruises to obtain Sargasso Sea water (see letter from Drs. Johnson and Bates). The water will be collected at the BATS site just before heading back to the dock at the Bermuda Institute of Ocean Sciences (BIOS). The estimated transit time is 8–9 hours. Upon arrival to BIOS, the sample will be processed and the experiments set up at laboratory space available for visiting scientist use (see letter from Dr. Blanco-Bercial).

Quantification of isotopic enrichment and trophic elemental transfer (NanoSIMS). Isotope labeled samples will be analyzed via nanoSIMS as previously published (Arandia-Gorostidi et al., 2017; preprint-Martínez Martínez et al., 2024). Briefly, incubated samples will be fixed with formaldehyde and filtered onto 0.2 µm pore size polycarbonate filters and gold coated. Samples will be pre-imaged with scanning electron microscope to find protist cells, and those protist cells and surrounding bacteria will be analyzed with the LLNL nanoSIMS for isotope quantification. Samples taken at time point zero will be used to calculate the fraction of protist and bacterial C and N incorporated into biomass as previously described (Dekas et al., 2019).

Taxonomic analyses. We will filter 750 ml of seawater from the <40 µm and < 0.8 µm environmental community fractions onto 0.22 µm at the beginning of the grazing experiments, before splitting into the treatment flask, and at the end of the experiments, from each of the corresponding treatment flasks, for DNA extraction and amplicon sequencing to determine changes in the protistan and bacterial communities. DNA will be extracted using commercially available kits (e.g., DNeasy® PowerWater, Qiagen). The 18S rRNA v4 hypervariable region and 16S rRNA v3-v4 hypervariable regions will be amplified, sequenced and analyzed to investigate the protistan and bacterial members, respectively, following procedures and bioinformatic previously described methods and pipelines (Blanco-Bercial et al., 2022; Zemb et al., 2020). For one experiment at each site and season, we will carry out stable isotope probing combined with microarray (Chip-SIP) (Mayali et al., 2014, 2012; Mayali and Weber, 2018). This method extracts isotope labeled RNA, hybridizes it to a high-density microarray targeting the 16S and 18S rRNA genes from taxa of interest, and analyzes the isotope composition of the array probe spots by nanoSIMS. We already designed a microarray targeting prokaryotes and eukaryotes from a previous experiment where EhV were added to seawater from the Bigelow Laboratory dock, and we will further refine the array with newly collected 16S and 18S rRNA gene amplicon data from the proposed experiments. The resulting data will provide relative isotope enrichment data (unlike the absolute enrichment data from the direct nanoSIMS analysis of cells), in order to directly identify the bacteria and protists consuming the virus particles.

Statistical analyses. The infection and grazing experiments will include biological triplicates. Furthermore, two independent grazing experiments (1–2 weeks apart) will be done at each site and season to test reproducibility and/or short terms variations. We will use R Statistical Software (latest version available at the time, R Core Team) for data analyses, exploration, and visualization. Optimal statistical analyses will be selected based on data distributions and homogeneity of variance. NanoSIMS analysis will be carried out on single replicates and the single cell incorporation data will be compared across samples and treatment with non-parametric analyses, including measures of heterogeneity (coefficient of dispersion). Biovolume and cell abundance data will be combined with cell-specific activity data to calculate volumetric flux of viral C and N into protist and bacterial size classes over time.

Model fitting procedures. We will use the Metropolis algorithm (Metropolis et al., 1953) to fit non-linear ODE models (e.g., Equation 1) to our experimental datasets. Co-PI Talmy has utilized the Metropolis algorithm to fit ODE models to laboratory data in the context of phytoplankton growth (Omta et al., 2020) and algal host-virus interactions (Hinson et al., 2023a). A Python package implementing a version of the Metropolis algorithm developed by the Talmy lab at UTK, available through a fully public GitHub repository, will be used for all model fitting.

GUIDING RESEARCH TIMELINE AND PROJECT RESPONSIBILITIES

Activity	2025				2026					2027					2028
	Mar-May	Jun-Aug	Sep-Nov	Dec-	-Feb	Mar-May	Jun-Aug	Sep-Nov	Dec-	-Feb	Mar-May	Jun-Aug	Sep-Nov	Dec-	-Feb
Postdoc and graduate student recruitment															
Set-up stock host and virus cultures															
Grazing experiments															
Design and virus stock labelling*															
Host and virus abundance - cytometry			Bigelow			Bigelow		BIOS			BIOS				
nanoSIMS and SEM**															
Taxonomic analyses**															
Modeling results*															
Data interpretation/manuscript preparation															
Broader impacts															
Mentoring students and postdoc															
Public outreach programs															

*Mathematical modeling is integrated throughout the experimental design and analyses; **LNLL sample analysis, postdoc-led data analysis (with advice from Dr. Mayali)

All team members will discuss the labeling and grazing experiments to ensure optimal design with full integration of mathematical modeling to facilitate implementation and results interpretation. We are requesting full-time support for one postdoc and one graduate student. Consequently, the postdoc will be in charge of setting up the laboratory experiments, collecting samples, analyzing the data, and leading the preparation of some manuscripts. The graduate student will work on the modeling aspects of the project and lead manuscripts. The PIs (part-time supported in this project) will teach, demonstrate, offer suggestions, and oversee the work of the junior researchers. Particularly, Martínez Martínez (Bigelow Laboratory) will lead the experimental virus/grazing work, Mayali (Adjunct Principal Investigator, Bigelow Laboratory), will lead the isotope probing analyses, and Talmy (UTK) will lead the work on mathematical modeling. Additionally, Martínez Martínez will assist with sample collection, data analysis and interpretation during some of the more labor intense experiments (particularly the Fall ones when undergraduates interns are not involved). All team members will contribute to data and results dissemination in the form of publications and/or public outreach.

BROADER IMPACTS

The results of this project will largely improve our understanding of microbes and their viruses as vital components of microbial communities and of nutrients and energy transfer within the microbial food web and to higher trophic levels in aquatic environments. The quantitative and mechanistic knowledge gained through this study will be foundational for other scientists to develop new hypotheses in subdisciplines across biological and chemical oceanography and microbial ecology and biology. The usefulness of our data products will be maximized by making all resources (data, protocols, and tools) readily publicly available through the Biological and Chemical Oceanography Data Management Office (BCO-DMO), *protocol.io*, and GitHub, as well as through peer-reviewed publications.

Education, mentoring, and outreach activities will be a significant component of this project. We will build upon and expand ongoing collaborations with highly successful programs to introduce early-career scientists, students, and the general public to scientific discoveries and hands-on training. The success of these activities will be evaluated through mechanisms established at the PIs' institutions that consider the number and demographics of participants and attendees to public events, feedback on evaluation forms, scientific outcomes (e.g., number of publications, open modeling code in GitHub repositories, number of presentations at scientific meetings, etc), career progression of all team members, and retention in STEM.

The PIs' multidisciplinary expertise will offer exceptional cross-training opportunities for a **postdoctoral researcher** at Bigelow Laboratory (to be hired), who will focus on the empirical aspects of the project, a **graduate student** at UTK, who will focus on mathematical modeling to support the bulk of their Ph.D dissertation, and 1 - 2 paid (with funds requested in this project) **undergraduate students** at both institutions each year. The trainees will actively participate in regular team meetings and designing and performing integrated laboratory and computing experiments, ensuring they have a voice and meaningful intellectual input on the project. The postdoc and the graduate student are expected to be the lead authors in publications that derive from this project. As part of their training, the postdoc and the graduate student will co-supervise undergraduates (who will also be co-authors in publications) and participate in outreach activities. Additionally, they will be invited to give seminars on their complementary projects at Bigelow Lab and UTK to foster networking within the collaborative team and other scientists. The postdoc will work with co-PI Mayali at LLNL to receive training in isotope-tracing techniques and data analyses. This project presents a rare opportunity for the students and the postdoc to collaborate with experimentalists and modelers across institutions on the same project to integrate biology, chemistry, and mathematics, providing an invaluable experience for their future careers working in interdisciplinary teams. This proposal's many self-contained sub-projects (e.g., viral infections, protist growth, and grazing rates) will provide exceptional opportunities for students to receive research experience and training, from experimental design to communication skills. Students will learn culturing techniques, flow cytometry, and interpretation of nanoSIM data. We expect the students to support the main objectives of this proposal through participation in experiments while being allowed to design tangential experiments, based on their interests, that expand on this project's cope and may count toward senior theses. Reciprocal expectations, problems, success, and pertinent literature will be discussed during weekly meetings. Student participation at meetings and activities of professional societies will be supported. Also, students will be encouraged and guided on seeking travel awards for in-person participation. These training experiences will be strong given the combined expertise and significant outcomes of the three PIs as scientists and inclusive mentors for individuals of groups underrepresented in STEM.

The postdoctoral researcher will be hired through an open search using Bigelow's human resources and DEI protocols and best practices for advertising and vetting candidates. The job advertisement will include language to encourage applications from underserved and underrepresented individuals in STEM. We will be share it personally with colleagues at several Minority Serving Institutions who can distribute it within their departments and help identify strong minority candidates. Bigelow undergraduate interns enrollment will be done in collaboration with the Biological Sciences department at Southern Maine Community College (SMCC) (see letter). Each year, Martínez Martínez will present a seminar at SMCC on the topic of this proposal to inform and encourage potential applicants. For years, Bigelow and SMCC have collaborated to support funded hands-on experiences for undergraduates from diverse identities and socioeconomic backgrounds. Martínez Martínez has previously incorporated underrepresented and minority students from SMCC and other institutions in his research.

Talmy has an appointment with the National Institute for Modeling Biological Synthesis (NIMBioS). In collaboration with biologists, he provides cross-training for NSF REU (Research Experience for undergraduates), graduate, and postdoctoral scientists. Talmy has prior successes as an inclusive mentor for BIPOC and Latinx students through his work with the National Summer Undergraduate Research Project (NSURP) initiative. By continuing this work and through his laboratory's explicit commitment to diversity, he will ensure that he successfully recruits students to work on the proposed project.

Each summer of the project, Martínez Martínez will serve as faculty during the annual **SigmaCamp** (see letter of collaboration, food and lodging are provided). SigmaCamp, currently in its 13th year, started as a collaborative effort of a group of scientists with a passion for extracurricular education and a mission to inspire gifted students in STEM topics. Every year, 120 students (12–16 years old) participate in this weeklong math and science sleepaway camp to learn about complex scientific topics in the format of lectures and hands-on workshops and laboratories. Applications are need-blind and scholarships are available for all qualified students. The instructors include researchers from many institutions at different career stages (from undergraduate to professor). The academic program is meticulously designed to promote collaboration and highlight the interconnectedness of scientific disciplines across Math, Physics, Chemistry, Biology, Computer Science, and Engineering. Most of the former campers now hold jobs in STEM or are pursuing graduate degrees and have returned to SigmaCamp as faculty. This August, is the first time that Martínez Martínez participates in SigmaCamp. He will give a one-hour lecture titled “Marine viruses: TINY biological entities with GIANT global impacts” and co-teach a 5-day long hands-on laboratory (*semilab*) about microbial ecology and biological oceanography titled “The unseen world of the sea”. For the length of this project, the postdoctoral researcher will be invited to give an introductory lecture about aquatic virus ecology and Martínez Martínez will lead a new *semilab* that highlights the concept of viruses and food, where student teams will grow heterotrophic protists on varying amounts of virus concentrates. The students will learn how to enumerate protists using microscopy, plot simple growth curves, and compare and discuss their results. Martínez Martínez will guide discussions about the ecological implications of their study. Existing SigmaCamp assessment tools will be used to evaluate the effectiveness of the proposed new virovory content. Based on this feedback, the content of the lecture and the *semilab* will be refined in subsequent years.

Finally, Martínez Martínez and the postdoc will participate in Bigelow's annual Open House. These interactions provide an excellent opportunity to explain the importance of basic research and understanding marine ecology. Anecdotes and results from this study (viruses are predators and food) will provide valuable tools for engaging the general public.

RESULTS FROM PRIOR NSF SUPPORT

OPP-1644155 Viral control of microbial communities in Antarctic lakes; Aug. 1, 2017 – Jul. 31, 2022; $498,165. Martínez Martínez was the lead PI for this project until April 2019, when PI status was changed to Dr. Twining (Bigelow Lab). Martínez Martínez then served as an unfunded collaborator. **Intellectual Merit:** This project investigated the viral communities in Antarctic lakes and the Southern Ocean. New data from this project was deposited in the U.S. Antarctic Program Data Center (USAP-1644155_1). The project leveraged sequencing data available through JGI and NCBI. Presentations: Haggett et al. (2019), ASLO ASM; Cretoiu et al. (2020), ASLO OSM; Publications: (De Corte et al., 2019; Hawco et al., 2022; Martínez et al., 2020). **Broader Impacts:** The project provided scientific training in aquatic virology and computational biology for two female postdoctoral researchers and a first-generation community college, female undergraduate intern Emily Haggett. The intern received a travel award from Bigelow Laboratory to present her work at the 2019 ASLO ASM. Martínez Martínez collaborated with Maine-based artist Justin Levesque and discussed the discoveries from this work with non-scientific audiences through Bigelow Laboratory’s outreach events.

OCE-2023680 Characterizing the effects of exogenous reactive oxygen species on marine microbial ecosystem dynamics; Aug. 1, 2020 – July 31, 2024. Total $913,189. Talmy served as PI on OCE-2023680. **Intellectual Merit:** This award is supporting the development of a deeper and more predictive understanding of how microbial community composition and mortality depend upon ROS production and decay. **Broader Impacts:** This award involves the training of undergraduate and graduate students in oceanographic research and public outreach about microbiology and oceanography to the local Knoxville community, as well as dependents of active duty marines. This support has resulted in 11 peer-reviewed publications (Cael et al., 2021; Calfee et al., 2022; Garcia et al., 2022; Glasgo et al., 2024; Hinson et al., 2023b; Lennon et al., 2024; Martiny et al., 2022; Omta et al., 2023; Papoulis et al., 2021; Rajakaruna et al., 2022; Talmy et al., 2024), two Ph.D. dissertations and one Masters thesis thus far.